LDA Homework 1 (2025-2026)

Question 1

Check if Data is Balanced

From the description of the experiment we note that there is dropout during follow up. Before using any summary methods to check the mean, variance and correlation structure we first check if the data is balanced. We see that there is missingness in the data as shown in Figure @ref(fig:q1_missing_values). Patients with higher baseline BPRS scores tend to miss more follow-up visits. There is also an increase of of the percentage of missingness over time, reaching above 40% by year 6. Because simple by time averages overweight those who remain and under represent those who dropped out (patients with higher baseline BPRS scores), they can paint a biased trend if simple bprs by time averages are used. This pattern suggests data are not missing completely at random, so we use smoothing methods that are better suited for unbalanced data

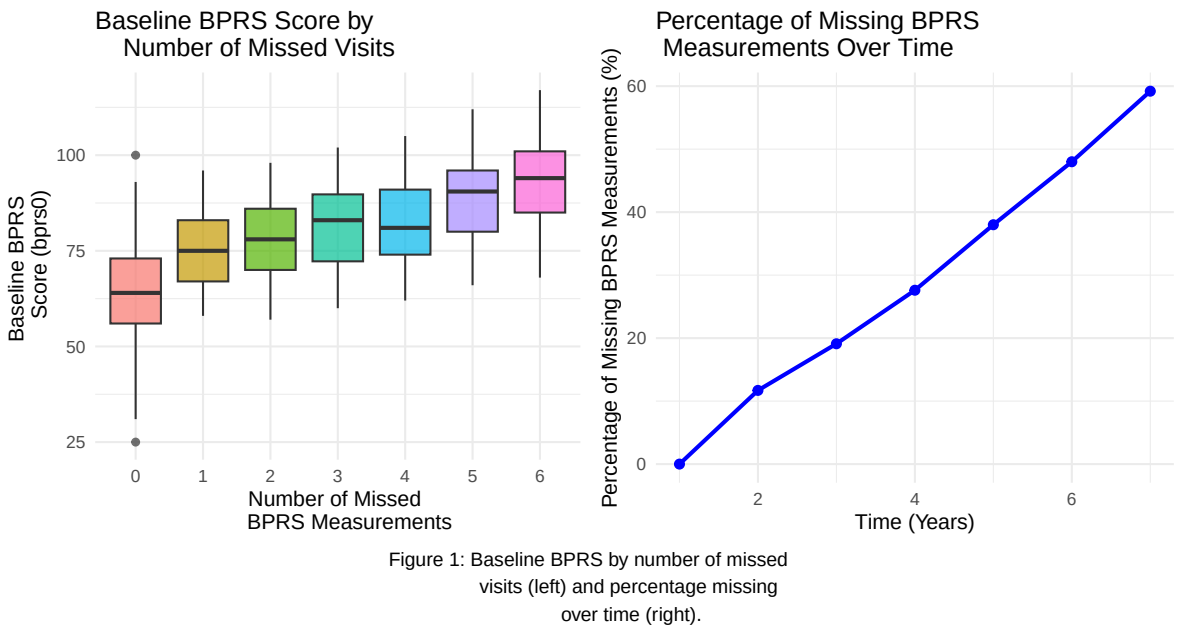

Figure 1: Baseline BPRS by number of missed visits (left) and percentage missing over time (right).

Zero-variance check for baseline Tau PET (taupet0)

From the table, taupet0 still varies through Year 2, but from Year 3 onwards it has zero variance (only one unique value among those with BPRS observed). That means taupet0 cannot explain any between-patient differences after Year 2 and coefficients for those years are not estimable. We should not use taupet0 from year 3 as this variable would not provide information on growth trajectories from here onwards.

Table 1: Baseline taupet0 variability among patients with observed BPRS at each year

n_patients	n_unique_taupet0	n_unique_abet0	sd_taupet0	sd_abet0	year	zero_variance
1253	27	67	0.1159765	0.4535830	0	FALSE
1106	13	65	0.0492840	0.4208376	1	FALSE
1014	6	62	0.0231188	0.3967379	2	FALSE
907	1	55	0.0000000	0.3730319	3	TRUE
777	1	50	0.0000000	0.3245475	4	TRUE
652	1	39	0.0000000	0.2718052	5	TRUE
511	1	22	0.0000000	0.2131188	6	TRUE

Average loess smoothing

Categorical Variables

To explore potential subgroup differences and to see how various variables are associated with the evolution of BPRS, we create LOESS smoothed plots of BPRS over time stratified by categorical variables: sex, education, job status, living situation, and clinical center. Across these variables, BPRS increases over time. Most subgroup curves are roughly parallel, showing main effects without interactions for variables like sex, job status, and living situation. The four education categories overlap almost completely and there is no difference in trend. This means there is unlikely a significant effect of education on BPRS trajectory.

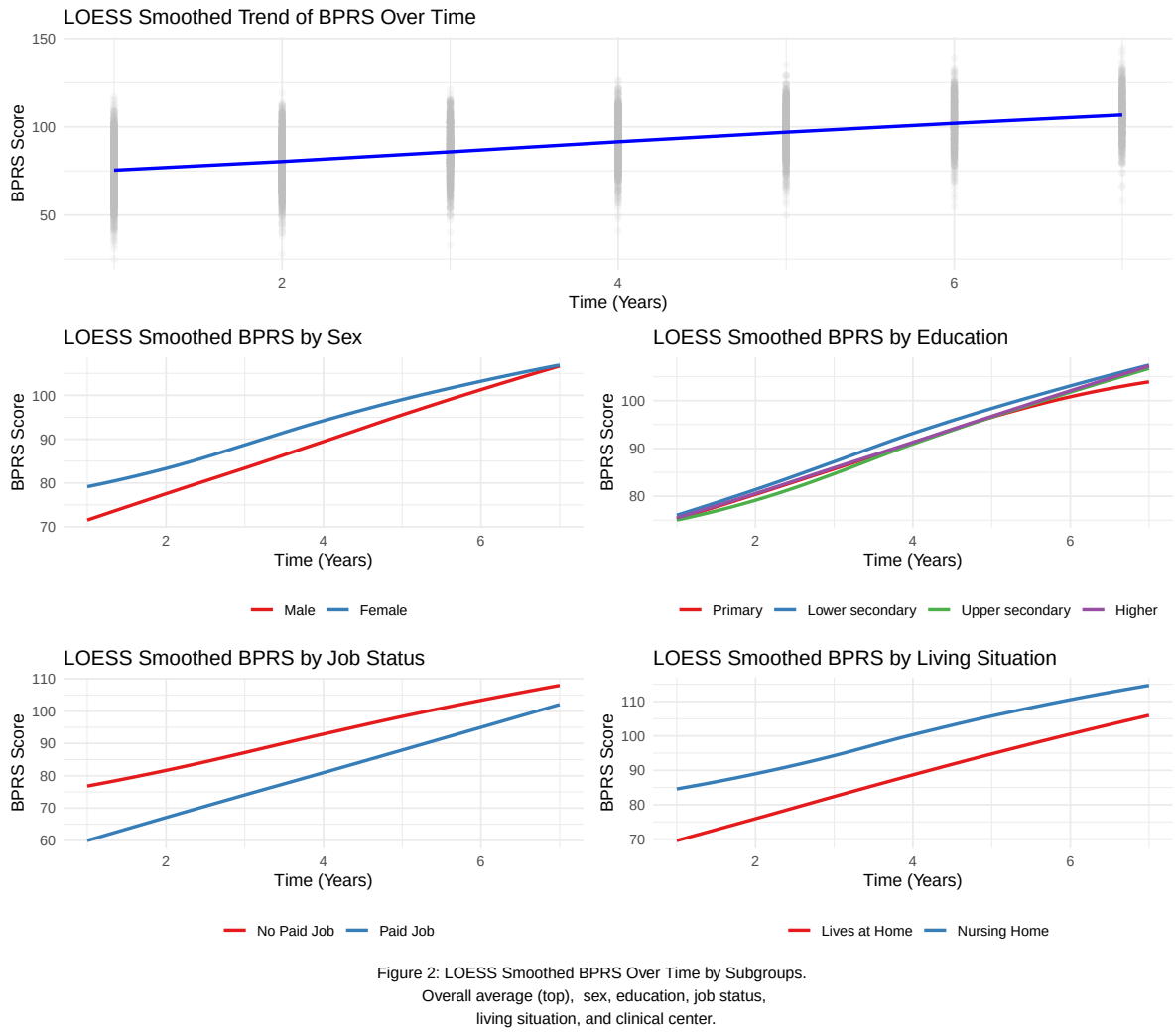

Figure 2: LOESS Smoothed BPRS Over Time by Subgroups: overall average (top), sex, education, job status, living situation, and clinical center.

Continuous Variables

To explore continuous predictors, we plotted LOESS-smoothed BPRS against age, BMI and baseline CDR-SB (cdrsb0) and abept0 using separate curves for each follow-up year. Higher age is associated with higher BPRS at every visit. BMI shows a curved pattern that peaks in the mid-20s, suggesting that a non-linear term would describe it better than a straight line. Baseline CDR-SB shows only a weak and slightly curved association with BPRS. The LOESS plot of BPRS versus ABPET0 by visit year shows an unstable curved pattern, with peak at

year 3 that is not repeated at other visits, which might be because of high missingness of this variable in follow up years

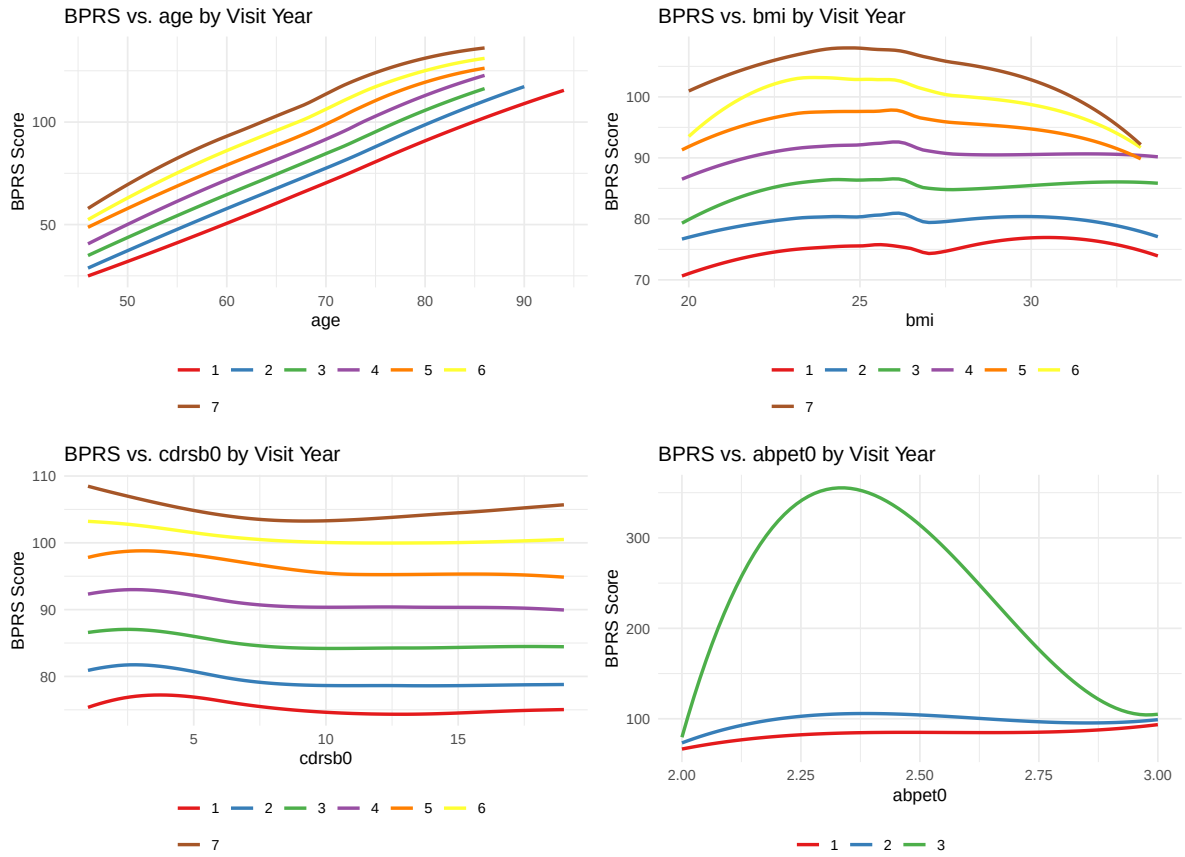

Figure 3: LOESS Smoothed BPRS over continuous predictors by visit year: age, BMI, baseline CDR-SB (cdrsb0), and baseline ABPET0.

Trial Centers Plot

To check whether clinical center (trial) affects the mean structure, we plot (Figure @ref(fig:q1_trial_centers)) LOESS-smoothed BPRS over time separately by trial centre. We see clear separate curves between centers, indicating that trial should be included as a random effect in the model to account for between center variability.

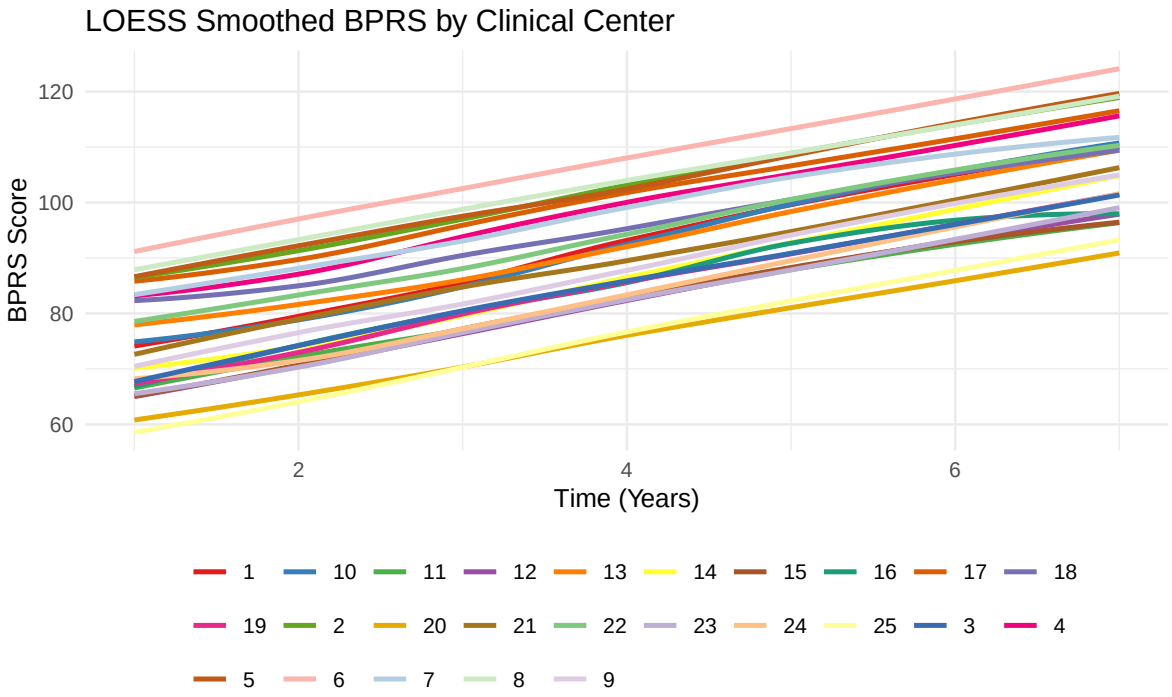

Figure 4: LOESS Smoothed BPRS Over Time by Clinical Center.

Figure 4: LOESS Smoothed BPRS Over Time by Clinical Center.

Variance & Correlation Structure

The smoothed squared residuals shows a flat residual curve, we dont observe a heteroscedastic pattern, meaning the spread of patients BPRS's readings trajectory around there expected readings doesn't change a lot. The semi-variogram also rises with the time gap between measurements, showing that observations from the same patient are more similar when close together and become less similar as the gap widens. These pattern indicates a within-patient correlation that weakens with distance in time therefore, the models we fit need to allow a correlation structure that decays over time. The plots are shown in Figure @ref(fig:q1_variance_correlation).

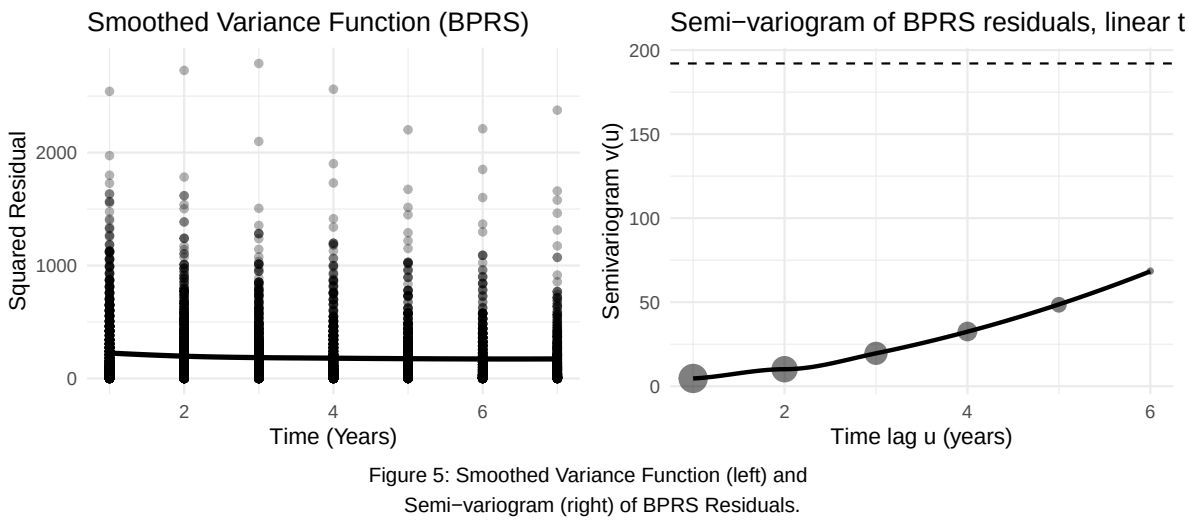

Figure 5: Smoothed variance function (left) and semivariogram (right) of BPRS residuals.

Question 1: Conclusions and modeling implications

- Mean structure: BPRS rises steadily over time. Subgroup curves are mostly parallel with level shifts by sex, job, living situation, and clear vertical shifts by clinical center; education shows little separation.
- For continuous variables age, showed a straight line pattern while BMI showed non linear relationship with BPRS.
- Variance structure: The smoothed squared-residual function is roughly flat across visits (no visual heteroscedasticity observed). We will fit a model with un assumed heterogeneous variance and compare this with a constant variance model using AIC.
- Correlation structure: The semivariogram increases with time lag, indicating strong within-patient correlation at short lags that decays as visits are farther apart.
- Missingness: Dropout increases over time and is more common for patients with higher baseline BPRS, so simple time-point averages would be biased
- Modeling implications: For population-average inference, we will fit repeated measures model with unstructured covariance structure and compare with AR(1) correlation
- For subject-specific inference, we will fit a mixed model with patient random intercept and slope and include a random intercept for clinical center, because there is a clear clinical site differences
- we don't recommend using taupet 0 as a covariate in longitudinal models as it has zero variance from year 3 onwards, for the participants who's BPRS measurement were taken at year 3 and beyond. Hence it will not show how taupet0 affects BPRS over the years

Question 2

Individual profiles & per-patient slopes

To visualise heterogeneity in longitudinal evolution, BPRS trajectories of 40 random patients were plotted. These individual profiles show variation in baseline symptom severity, together with increasing trends over time. The profiles also illustrate the unbalanced nature of the data, as subjects contribute different numbers of observations. These observations help justify the choice of summary statistics for longitudinal data.

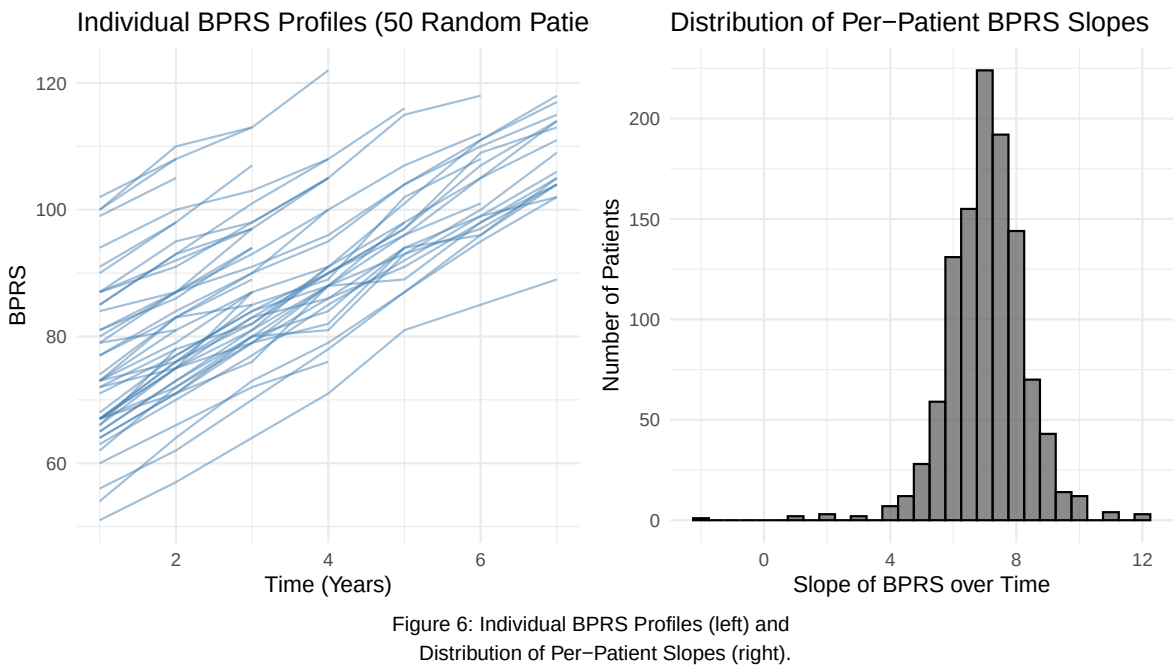

Figure 6: Individual BPRS profiles (left) and distribution of per-patient slopes (right).

Separate analyses for each follow-up year

For each follow-up year $t \in \{1, \dots, 6\}$, we fit an ordinary linear model relating the BPRS score at year t to baseline covariates:

$$\begin{aligned} extBPRS_{it} = \alpha_{0t} + \alpha_{1t} \cdot \text{sex}_i + \alpha_{2t} \cdot \text{age}_i + \alpha_{3t} \cdot \text{bmi}_i + \alpha_{4t} \cdot \text{job}_i + \alpha_{5t} \cdot \text{adl}_i \\ + \alpha_{6t} \cdot \text{wzc}_i + \alpha_{7t} \cdot \text{cdrsb0}_i + \alpha_{8t} \cdot \text{abpet0}_i + \alpha_{9t} \cdot \text{taupet0}_i + \varepsilon_{it}. \end{aligned}$$

where, for each fixed year t:

- α_{0t} is the expected BPRS at year t for the reference patient (Male, No Paid Job, Lives at Home), with continuous covariates at 0.
- α_{1t} is the difference at year t for Female vs Male.
- α_{2t} is the change at year t per 1-year increase in age.
- α_{3t} is the change at year t per 1-unit increase in BMI.
- α_{4t} is the difference at year t for Paid Job vs No Paid Job.
- α_{5t} is the change at year t per 1-unit increase in ADL.
- α_{6t} is the difference at year t for Nursing Home vs Lives at Home.
- α_{7t} is the change at year t per 1-unit increase in baseline CDR-SB (cdrsb0).
- α_{8t} is the change at year t per 1-unit increase in baseline Amyloid PET (abpet0).
- α_{9t} is the change at year t per 1-unit increase in baseline Tau PET (taupet0).
- ε_{it} is the error term at year t.

Table 2: Separate Linear Models for Each Follow-up Year

covariate	Year0	Year1	Year2	Year3	Year4	Year5	Year6
(Intercept)	-122.72 [-126.28, -119.16]	-119.78 [-126.03, -113.52]	-105.57 [-120.15, -90.99]	-104.92 [-111.33, -98.51]	-101.46 [-109.63, -93.29]	-95.55 [-106.07, -85.02]	-87.73 [-101.67, -73.79]
abpet0	0.34 [-0.07, 0.74]	0.20 [-0.29, 0.69]	0.03 [-0.61, 0.68]	-0.13 [-0.98, 0.71]	-0.03 [-1.18, 1.13]	-0.10 [-1.74, 1.54]	-1.11 [-3.75, 1.54]
adl	2.01 [1.92, 2.11]	2.17 [2.05, 2.29]	2.25 [2.09, 2.41]	2.11 [1.91, 2.32]	2.26 [2.00, 2.52]	2.37 [2.02, 2.71]	2.42 [1.96, 2.87]
age	2.36 [2.33, 2.40]	2.41 [2.36, 2.45]	2.42 [2.37, 2.48]	2.41 [2.34, 2.48]	2.45 [2.36, 2.54]	2.48 [2.37, 2.58]	2.50 [2.37, 2.64]
bmi	0.49 [0.44, 0.55]	0.52 [0.45, 0.58]	0.51 [0.42, 0.59]	0.54 [0.43, 0.65]	0.56 [0.42, 0.70]	0.51 [0.33, 0.69]	0.47 [0.23, 0.71]
cdrsb0	-0.02 [-0.04, -0.01]	-0.04 [-0.06, -0.02]	-0.06 [-0.08, -0.04]	-0.08 [-0.11, -0.05]	-0.13 [-0.17, -0.09]	-0.15 [-0.20, -0.10]	-0.17 [-0.23, -0.10]
jobPaid Job	-18.33 [-19.09, -17.57]	-19.24 [-20.17, -18.30]	-19.77 [-20.97, -18.56]	-19.01 [-20.54, -17.47]	-20.19 [-22.13, -18.25]	-20.86 [-23.30, -18.41]	-21.07 [-24.18, -17.96]
sexFemale	-0.28 [-0.51, -0.05]	-0.40 [-0.68, -0.12]	-0.18 [-0.54, 0.18]	-0.22 [-0.68, 0.25]	-0.34 [-0.94, 0.26]	-0.37 [-1.15, 0.41]	-0.57 [-1.64, 0.49]
taupet0	0.43 [-0.57, 1.43]	0.37 [-2.34, 3.09]	-4.06 [-11.47, 3.35]	NA	NA	NA	NA
wzcNursing Home	1.69 [1.43, 1.94]	1.94 [1.63, 2.25]	2.10 [1.69, 2.51]	2.24 [1.69, 2.80]	1.94 [1.18, 2.71]	1.43 [0.34, 2.52]	2.16 [0.42, 3.90]

Question 3

Motivation Repeated-Measures Model

In Question 2, we analysed each year separately and presented yearly results (Table @ref(q2_separate_year_models)). While this approach is simple, it has limitations. Running separate models for each year means repeating the same hypothesis test multiple times, which increases the risk of Type I error. It also reduces statistical power, because each model uses only one year of data instead of leveraging all observations together. Additionally, separate analyses do not address the main research question: how BPRS scores changes over time and how baseline characteristics influence that change. They also ignore the fact that repeated measurements from the same patient are correlated, which can lead to misleading conclusions.

To overcome these limitations when estimating the population effects, we use a repeated-measures multivariate model. This approach treats each patient's yearly BPRS readings dependent. A normal multivariate linear model assumes all observations are independent, which is not realistic here because the same patient BPRS measurements are taken yearly up to

year 6. The repeated measures model correctly handles the correlation within a person, the uneven follow-up due to dropout, and the changing variability across visits.This model uses all available data and gives more reliable averages and standard errors. We start from a saturated repeated-measures model where time and all baseline characteristics enter the mean structure, and time interactions are included for sex, age, BMI, job, ADL, living situation (wzc), baseline CDR-SB, and baseline Amyloid PET (abpet0). Education is included as a main effect but not interacted with time. This allows both baseline levels and changes over time to vary with key predictors so that we do not miss important effects when we later simplify the model. For the repeated measurements within a patient, we allow a very flexible pattern for how BPRS scores at different visits are correlated to each other, without forcing a specific correlation pattern in advance. We do not use this saturated model to report final clinical results; we use it as a screening model and then simplify the mean structure.

$$
\begin{aligned}
E[Y_{ij} \mid X_i] = {} & \beta_0 + \beta_1\,\mathrm{time}_j + \beta_2\,\mathrm{edu}_i + \beta_3\,\mathrm{sex}_i + \beta_4\,\mathrm{age}_i + \beta_5\,\mathrm{bmi}_i + \beta_6\,\mathrm{job}_i \\
& + \beta_7\,\mathrm{adl}_i + \beta_8\,\mathrm{wzc}_i + \beta_9\,\mathrm{cdrsb0}_i + \beta_{10}\,\mathrm{abpet0}_i \\
& + \beta_{11}\,(\mathrm{time}_j \times \mathrm{sex}_i) + \beta_{12}\,(\mathrm{time}_j \times \mathrm{age}_i) + \beta_{13}\,(\mathrm{time}_j \times \mathrm{bmi}_i) + \beta_{14}\,(\mathrm{time}_j \times \mathrm{job}_i) \\
& + \beta_{15}\,(\mathrm{time}_j \times \mathrm{adl}_i) + \beta_{16}\,(\mathrm{time}_j \times \mathrm{wzc}_i) + \beta_{17}\,(\mathrm{time}_j \times \mathrm{cdrsb0}_i) + \beta_{18}\,(\mathrm{time}_j \times \mathrm{abpet0}_i)
\end{aligned}
$$

where:

- Y_{ij} is the BPRS score for patient i at time j.
- β_0 is the intercept: expected BPRS at time 0 for the reference patient (Male, No Paid Job, Lives at Home), with continuous covariates at 0.
- β_1 is the main effect of time: average yearly change in BPRS for the reference patient.
- β_2 is the main effect of education.
- β_3 is the main effect of sex: Female vs Male at time 0.
- β_4 is the main effect of age: per 1-year increase at baseline.
- β_5 is the main effect of BMI: per 1-unit increase at baseline.
- β_6 is the main effect of job: Paid Job vs No Paid Job at time 0.
- β_7 is the main effect of ADL: per 1-unit increase at baseline.
- β_8 is the main effect of living situation (wzc): Nursing Home vs Lives at Home at time 0.
- β_9 is the main effect of baseline CDR-SB (cdrsb0): per 1-unit increase at baseline.
- β_{10} is the main effect of baseline Amyloid PET (abpet0): per 1-unit increase at baseline.
- $\beta_{11}, \ldots, \beta_{18}$ are the time-by-covariate interactions for sex, age, BMI, job, ADL, wzc, CDR-SB, and abpet0, respectively.

The yearly BPRS change (slope with respect to time) under this model is

Table 3: Gls Full Model Results

term	estimate	std.error	statistic	p.value
(Intercept)	-129.58 [-133.00, -126.15]	1.75	-74.20	<0.01
time	5.96 [4.27, 7.64]	0.86	6.91	<0.01
eduLower secondary	0.20 [-0.13, 0.52]	0.17	1.19	0.23
eduUpper secondary	0.12 [-0.19, 0.42]	0.16	0.75	0.46
eduHigher	0.19 [-0.11, 0.50]	0.16	1.23	0.22
sexFemale	-0.24 [-0.49, 0.00]	0.13	-1.95	0.05
age	2.36 [2.32, 2.40]	0.02	126.37	<0.01
bmi	0.51 [0.45, 0.57]	0.03	17.00	<0.01
jobPaid Job	-18.13 [-18.91, -17.35]	0.40	-45.69	<0.01
adl	2.00 [1.90, 2.10]	0.05	38.57	<0.01
wzcNursing Home	1.63 [1.35, 1.92]	0.15	11.16	<0.01
cdrsb0	0.00 [-0.01, 0.02]	0.01	0.21	0.83
abpet0	0.34 [-0.10, 0.78]	0.22	1.52	0.13
time:sexFemale	-0.03 [-0.16, 0.09]	0.06	-0.52	0.61
time:age	0.01 [-0.00, 0.03]	0.01	1.59	0.11
time:bmi	-0.00 [-0.03, 0.03]	0.01	-0.06	0.95
time:jobPaid Job	-0.39 [-0.78, 0.01]	0.20	-1.90	0.06
time:adl	0.05 [-0.00, 0.11]	0.03	1.95	0.05
time:wzcNursing Home	0.14 [-0.00, 0.29]	0.07	1.90	0.06
time:cdrsb0	-0.02 [-0.03, -0.01]	0.00	-5.10	<0.01
time:abpet0	-0.07 [-0.29, 0.16]	0.11	-0.58	0.56

Simplified Mean Structure

For this step we fit a repeated-measures model with a simplified mean structure but the same unstructured covariance as before. In this model, time and all main baseline characteristics except education remain in the mean structure, and only one time interaction (time with baseline CDR-SB) is kept. The estimated correlation matrix shows a positive correlation between repeated BPRS scores within a patient, especially for visits that are closer in time, and the visit-specific standard deviations increase from baseline to later years, contradicting with our earlier variance plots showing homoscedastic variance. We will use this reduced-mean, unstructured covariance model as the starting point when we compare with simpler covariance structures, for instance AR(1).

$$\begin{aligned} E[Y_{ij} \mid X_i] = \beta_0 &+ \beta_1 \,\mathrm{time}_j + \beta_2 \,\mathrm{sex}_i + \beta_3 \,\mathrm{age}_i + \beta_4 \,\mathrm{bmi}_i + \beta_5 \,\mathrm{job}_i \\ &+ \beta_6 \,\mathrm{adl}_i + \beta_7 \,\mathrm{wzc}_i + \beta_8 \,\mathrm{cdrsb0}_i \\ &+ \beta_9 \,(\mathrm{time}_j \times \mathrm{cdrsb0}_i) \end{aligned}$$

where:

- Y_{ij} is the BPRS score for patient i at time j.
- β_0 is the intercept: expected BPRS at time 0 for the reference patient (Male, No Paid Job, Lives at Home), with continuous covariates at 0.
- β_1 is the main effect of time: average yearly change in BPRS for the reference patient.
- β_2 is the main effect of sex: Female vs Male at time 0.
- β_3 is the main effect of age: per 1-year increase at baseline.
- β_4 is the main effect of BMI: per 1-unit increase at baseline.
- β_5 is the main effect of job: Paid Job vs No Paid Job at time 0.
- β_6 is the main effect of ADL: per 1-unit increase at baseline.
- β_7 is the main effect of living situation (wzc): Nursing Home vs Lives at Home at time 0.
- β_8 is the main effect of baseline CDR-SB (cdrsb0): per 1-unit increase at baseline.
- β_9 is the time-by-cdrsb0 interaction: modification of yearly change per 1-unit increase in baseline CDR-SB.

The covariance structure remains the same as in the full model.

The yearly BPRS change (slope with respect to time) in this simplified model is

$$\frac{\partial E[Y_{ij} \mid X_i]}{\partial \,\mathrm{time}_j} = \beta_1 + \beta_9 \,\mathrm{cdrsb0}_i.$$

Thus, for a 1-unit higher baseline CDR-SB, the yearly BPRS change increases by β_9 units, all else equal. When $\mathrm{cdrsb0}_i = 0$, the average yearly change is β_1.

Table 4: Simplfied mean structiure Unstructured Covariance Results

term	estimate	std.error	statistic	p.value
(Intercept)	-131.59 [-134.11, -129.08]	1.28	-102.43	<0.01
time	7.22 [7.14, 7.30]	0.04	179.38	<0.01
sexFemale	-0.27 [-0.46, -0.07]	0.10	-2.72	<0.01
age	2.39 [2.37, 2.42]	0.01	212.64	<0.01
bmi	0.51 [0.47, 0.55]	0.02	22.80	<0.01
jobPaid Job	-18.69 [-19.30, -18.08]	0.31	-59.80	<0.01
adl	2.08 [2.01, 2.16]	0.04	53.17	<0.01
wzcNursing Home	1.84 [1.62, 2.05]	0.11	16.86	<0.01
cdrsb0	0.00 [-0.01, 0.02]	0.01	0.35	0.73
time:cdrsb0	-0.02 [-0.03, -0.01]	0.00	-5.29	<0.01

LR test between full and reduced mean structure models

The likelihood ratio test compares the complex mean structure model (with all time–covariate interactions) to the reduced model (only time $\times$ CDR-SB). The test statistic is 13.78 with a p-value of 0.25, which means there is no evidence that the extra interaction terms improve the fit. This shows the simpler mean structure explains the data just as well as the saturated one, while using fewer parameters, so we will keep the reduced model as our working mean structure.

Table 5: Likelihood Ratio Test between Full and Reduced Mean Structure Models

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
gls_full_mean_structure	1	49	27318.29	27648.33	-13610.14		NA	NA
gls_reduced_mean_structure	2	38	27310.07	27566.02	-13617.03	1 vs 2	13.78	0.25

Reduced Covariance Structure

For this step, we keep the same simplified mean structure and only change how we model the correlation over time. The aim is to check whether a simpler covariance structure is sufficient. We fit a repeated-measures model with an AR(1) correlation structure, where each visit is highly correlated with the previous one and the correlation weakens as the time gap increases. We still allow variability to differ by visit. This means BPRS measurements taken close in time tend to be similar, and later visits show more spread.

When we compare this AR(1) model to the more flexible model with a unstructured covariance structure, the likelihood ratio test shows (L.Ratio ≈ 1233, $p < 0.0001$) and AIC/BIC worsen for AR(1). This indicates that the data pattern is too complex to be captured by this AR(1)

correlation structure, so for population-average inference we keep the unstructured covariance from the previous model.

Table 6: Marginal Model with AR(1) Covariance Results

term	estimate	std.error	statistic	p.value
(Intercept)	-131.98 [-135.20, -128.75]	1.64	-80.27	<0.01
time	7.24 [7.17, 7.31]	0.03	208.57	<0.01
sexFemale	-0.28 [-0.53, -0.03]	0.13	-2.24	0.03
age	2.40 [2.37, 2.43]	0.01	166.48	<0.01
bmi	0.50 [0.45, 0.56]	0.03	17.57	<0.01
jobPaid Job	-18.91 [-19.69, -18.12]	0.40	-47.07	<0.01
adl	2.10 [2.00, 2.20]	0.05	41.83	<0.01
wzcNursing Home	1.84 [1.57, 2.12]	0.14	13.25	<0.01
cdrsb0	0.00 [-0.02, 0.03]	0.01	0.33	0.74
time:cdrsb0	-0.02 [-0.03, -0.02]	0.00	-6.67	<0.01

Reduced mean structure vs reduced covariance structure

Table 7: Likelihood Ratio Test between Reduced Mean Structure and Reduced Covariance Structure Models

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
gls_reduced_mean_structure	1	38	27310.07	27566.02	-13617.03		NA	NA
gls_reduced_cov_structure	2	18	28503.30	28624.54	-14233.65	1 vs 2	1233.23	0

Discussion of Question 3 Results

For Question 3, our final repeated-measures model uses a reduced mean structure with a unstructured covariance. On the mean side, BPRS increases on average by about 7 points per year. After adjustment, higher age, higher BMI, worse ADL, job status, and living in a nursing home are all associated with higher BPRS levels, while women have slightly lower BPRS than men. The interaction of time and CDR-SB term is is significant meaning patients who start with better cognition tend to show a faster increase in BPRS, whereas those who already have worse cognition at baseline worsen more slowly. This directly answers our main question at the population level: BPRS worsens over time, and the speed of worsening is associated with baseline cognitive status.

For the biomarkers, we focus on ABPET0 and CDR-SB. ABPET0 was not retained in the reduced marginal model after simplification. The LOESS plot of BPRS versus ABPET0 by visit year shows an unstable curved pattern, with a peak at year 3 that is not repeated at

other visits, and its association with BPRS is uncertain. TAUPET0, on the other hand, was dropped because from year 3 onwards it has no variability, so it cannot explain differences in BPRS.

The covariance part of the model lets each visit have its own variability and allows all time points within a patient to be correlated without imposing a correlation structure. This matches what we saw in the exploratory plots repeated BPRS measurements from the same patient are strongly correlated, especially when visits are close in time, and the spread increases over follow-up. Using this unstructured structure gives more realistic standard errors for the population averages.

Finally, this marginal model has some limitations compared with a mixed-effects model. It provides population-average effects but not patient-specific trajectories, and it does not explicitly model differences between clinical centres. The LOESS plot by centre shows centres with roughly parallel but clearly non overlapping curves for majority of centres, suggesting centre level differences in BPRS levels over time. In the current model, this extra variation is absorbed into the residual covariance rather than being separated out. In Question 5 we therefore move to a mixed-effects model with random intercepts and slopes for patients, and a random intercept for trial, to better describe individual and centre-specific disease trajectories while still addressing the main objective of how BPRS changes over time, and how those changes relate to baseline clinical characteristics, cognition, and biomarkers.

Question 4

In the first stage of a two-stage analysis, we fit a separate linear regression model for each patient to estimate their individual intercept and slope over time. For each patient i, the model is:

$$Y_{ij} = \beta_{0i} + \beta_{1i} \cdot \text{time}_j + e_{ij}$$

where:

- Y_{ij} is the BPRS score for patient i at time j.
- β_{0i} is the patient-specific intercept (estimated baseline BPRS).
- β_{1i} is the patient-specific slope (estimated rate of change in BPRS).
- e_{ij} are the individual error terms, assumed to be independent with mean zero.

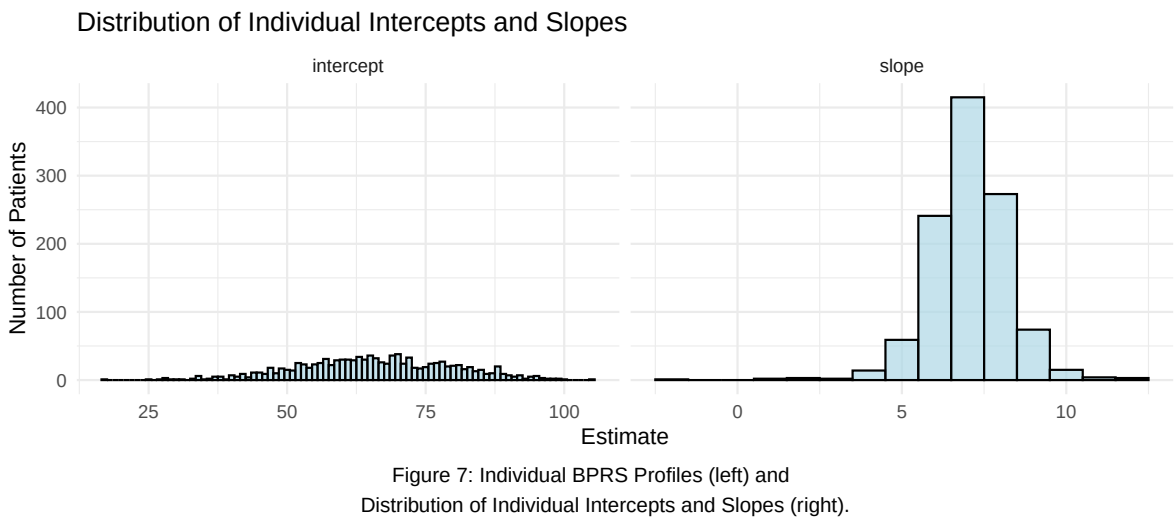

Figure 7: Distribution of individual intercepts and slopes from per-patient linear models.

Stage 2: Relate Intercepts and Slopes to Covariates

Model for the Intercepts

In the second stage, we model the estimated patient-specific intercepts ($\hat{\beta}_{0i}$) as a function of baseline covariates to understand which factors predict the initial BPRS score. The model is:

$$\begin{aligned}\hat{\beta}_{0i} =& \gamma_{00} + \gamma_{01} \cdot \text{sex}_i + \gamma_{02} \cdot \text{age}_i + \gamma_{03} \cdot \text{bmi}_i + \gamma_{04} \cdot \text{job}_i + \gamma_{05} \cdot \text{adl}_i \\ &+ \gamma_{06} \cdot \text{wzc}_i + \gamma_{07} \cdot \text{cdrsb0}_i + \gamma_{08} \cdot \text{abpet0}_i + \gamma_{09} \cdot \text{taupet0}_i + \epsilon_{0i}\end{aligned}$$

where:

- $\hat{\beta}_{0i}$ is the estimated intercept for patient i from Stage 1.
- γ_{00} is the average intercept for the reference individual (Male, No Paid Job, Lives at Home), with continuous covariates at 0.
- γ_{01} is the intercept difference for Female vs Male.
- γ_{02} is the change at baseline per 1-year increase in age.
- γ_{03} is the change at baseline per 1-unit increase in BMI.
- γ_{04} is the intercept difference for Paid Job vs No Paid Job.
- γ_{05} is the change at baseline per 1-unit increase in ADL.
- γ_{06} is the intercept difference for Nursing Home vs Lives at Home.
- γ_{07} is the change at baseline per 1-unit increase in baseline CDR-SB (cdrsb0).

- γ_{08} is the change at baseline per 1-unit increase in baseline Amyloid PET (abpet0).
- γ_{09} is the change at baseline per 1-unit increase in baseline Tau PET (taupet0).
- ϵ_{0i} is the error term.

Table 8: Two-Stage Intercept Model Results

term	estimate	std.error	statistic	p.value
(Intercept)	-130.85 [-137.31, -124.39]	3.29	-39.76	<0.01
sexFemale	-0.22 [-0.51, 0.06]	0.15	-1.52	0.13
age	2.36 [2.32, 2.41]	0.02	101.88	<0.01
bmi	0.51 [0.45, 0.58]	0.03	15.07	<0.01
jobPaid Job	-18.28 [-19.25, -17.32]	0.49	-37.15	<0.01
adl	2.03 [1.90, 2.15]	0.06	31.98	<0.01
wzcNursing Home	1.54 [1.22, 1.86]	0.16	9.41	<0.01
cdrsb0	0.00 [-0.02, 0.02]	0.01	0.12	0.90
abpet0	0.35 [-0.16, 0.86]	0.26	1.35	0.18
taupet0	0.53 [-2.27, 3.33]	1.43	0.37	0.71

Model for the Slopes

Similarly, we model the estimated patient-specific slopes ($\hat{\beta}_{1i}$) as a function of baseline covariates to see what predicts the rate of change in BPRS. The model is:

$$\begin{aligned}\hat{\beta}_{1i} =& \gamma_{10} + \gamma_{11} \cdot \text{sex}_i + \gamma_{12} \cdot \text{age}_i + \gamma_{13} \cdot \text{bmi}_i + \gamma_{14} \cdot \text{job}_i + \gamma_{15} \cdot \text{adl}_i \\ &+ \gamma_{16} \cdot \text{wzc}_i + \gamma_{17} \cdot \text{cdrsb0}_i + \gamma_{18} \cdot \text{abpet0}_i + \gamma_{19} \cdot \text{taupet0}_i + \epsilon_{1i}\end{aligned}$$

where:

- $\hat{\beta}_{1i}$ is the estimated slope for patient i from Stage 1.
- γ_{10} is the average slope for the reference individual (Male, No Paid Job, Lives at Home), with continuous covariates at 0.
- γ_{11} is the slope difference for Female vs Male.
- γ_{12} is the slope change per 1-year increase in age.
- γ_{13} is the slope change per 1-unit increase in BMI.
- γ_{14} is the slope difference for Paid Job vs No Paid Job.
- γ_{15} is the slope change per 1-unit increase in ADL.
- γ_{16} is the slope difference for Nursing Home vs Lives at Home.
- γ_{17} is the slope change per 1-unit increase in baseline CDR-SB (cdrsb0).

- γ_{18} is the slope change per 1-unit increase in baseline Amyloid PET (abpet0).
- γ_{19} is the slope change per 1-unit increase in baseline Tau PET (taupet0).
- ϵ_{1i} is the error term.

Table 9: Two-Stage Slope Model Results

term	estimate	std.error	statistic	p.value
(Intercept)	6.37 [2.97, 9.77]	1.73	3.68	<0.01
sexFemale	-0.04 [-0.19, 0.11]	0.08	-0.55	0.58
age	0.01 [-0.01, 0.04]	0.01	1.02	0.31
bmi	0.00 [-0.03, 0.04]	0.02	0.09	0.93
jobPaid Job	-0.34 [-0.85, 0.17]	0.26	-1.32	0.19
adl	0.05 [-0.02, 0.11]	0.03	1.37	0.17
wzcNursing Home	0.19 [0.02, 0.36]	0.09	2.23	0.03
cdrsb0	-0.02 [-0.03, -0.01]	0.01	-4.30	<0.01
abpet0	-0.07 [-0.34, 0.20]	0.14	-0.53	0.60
taupet0	-0.13 [-1.61, 1.34]	0.75	-0.17	0.86

Question 5

Plausible random effects model

To understand how psychiatric symptoms evolve over time in Alzheimer's disease, we developed a statistical model that accounts for repeated measurements in each patient. This model allows each patient to have their own starting level of symptoms and their own rate of change over time. The outcome of interest was the Brief Psychiatric Rating Scale (BPRS), measured annually over six years.

We model the longitudinal BPRS scores using a linear mixed-effects model with random effects at both the patient and trial level:

$$\text{BPRS}_{it} = (\beta_0 + b_{0i} + u_{0j}) + (\beta_1 + b_{1i})\,\text{time}_{it} + \beta_2\text{sex}_i + \beta_3\text{age}_i + \beta_4\text{bmi}_i + \beta_5\text{job}_i + \beta_6\text{adl}_i + \beta_7\text{wzc}_i + \beta_8\text{cdrsb0}$$

where:

- (BPRS_{it}) is the BPRS score for patient (i) in trial (j) at time (t)

- (_0, ..., _9) are **fixed (population-level) effects**
- (b_{0i}) is the patient-specific random intercept
- (b_{1i}) is the **patient-specific random slope** for time
- (u_{0j}) is the **trial-specific random intercept**
- (_{it}) is the **residual error term**

The patient-level random effects are assumed to follow a bivariate normal distribution:

$$\begin{pmatrix} b_{0i} \\ b_{1i} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_{b0}^2 & \sigma_{b0,b1} \\ \sigma_{b0,b1} & \sigma_{b1}^2 \end{pmatrix} \right),$$

and the trial-level random intercept follows:

$$u_{0j} \sim N(0, \sigma_{\text{trial}}^2).$$

Model Motivation and Justification

Here, We chose to fit linear mixed effects model because our dataset contains repeated BPRS measurements for the same patients over 6 years and across multiple clinical trials, with clear variability both in baseline symptom severity and rate of change between patients. An unstructured covariance matrix was used for the random effects, allowing the intercept and slope to be correlated. The model included:

1) Random intercepts and random slopes for each patient, reflecting that individuals differ both in their baseline psychiatric symptom severity and their rate of change over time.
2) A random intercept for trial was added to account for systematic differences across the 25 clinical trials.
3) Fixed effects included all clinically relevant baseline covariates (age, sex, BMI, job status, ADL, nursing home residence (WZC), baseline CDR-SB, amyloid PET), ensuring adjustment for major patient characteristics.
4) An interaction term between time and CDRSB was included to assess whether cognitive impairment at baseline affects the rate of psychiatric symptom progression.

Estimation Method

The model was fitted using Restricted Maximum Likelihood (REML), which provides more reliable and less biased estimates of variance components, especially because our dataset had many clusters (patients, trials). Because our model includes random intercepts for both patients and trials, REML is appropriate to ensure stable estimates of these random effect variances.

Assumptions made

1) Linearity: BPRS changes linearly over the follow-up period.
2) Normality: Random effects and residual errors are approximately normally distributed.
3) Independence: After accounting for random effects, residuals are independent.
4) Homoscedasticity: The residual errors have constant variance across time and subjects.
5) Missing data (dropout) are at least missing at random (MAR) conditional on the covariates and random effects.
6) Correct Covariance Structure: The specified random-effects structure (random intercepts and slopes) appropriately captures within-subject and within-trial correlation.

Limitations

1) More complex to interpret for non-statisticians.The presence of fixed effects, random effects, and correlation structures makes interpretation harder compared to ordinary regression
2) Results depend on correct specification of random-effects structure. Misspecifying random intercepts/slopes or the covariance structure can lead to biased estimates or incorrect standard errors.
3) Convergence issues can arise with complex models. Models with multiple random effects, unstructured covariance matrices, or small sample sizes may fail to converge or produce unreliable estimates.
4) Assumptions such as normality of random effects and linearity may not fully hold in real data. Although LME models are robust, violation of assumptions especially the normality of random effects can affect estimates, particularly with small sample sizes.
5) MAR assumption cannot be fully verified.The model assumes missing data are Missing At Random, but this assumption is untestable and can lead to biased results if dropout relates to unobserved outcomes.

Diagnostics: Evaluating Model Appropriateness

To ensure that the linear mixed-effects model is statistically reliable, several diagnostic checks were performed. These checks help determine whether the model's assumptions are met, and whether the results are reliable.

Normality of Residuals (Q–Q Plot) Figure @ref(fig:q5_residual_diagnostics) (right) presents the Normal Q–Q plot which compares the distribution of the model residuals (errors) to what would be expected if they followed a normal distribution, which is an underlying assumption of linear mixed-effects models. The residuals aligned closely with the expected diagonal line, showing two deviations at each tail. The assumption of normally distributed residuals is largely satisfied. This supports the reliability of the estimated effects.

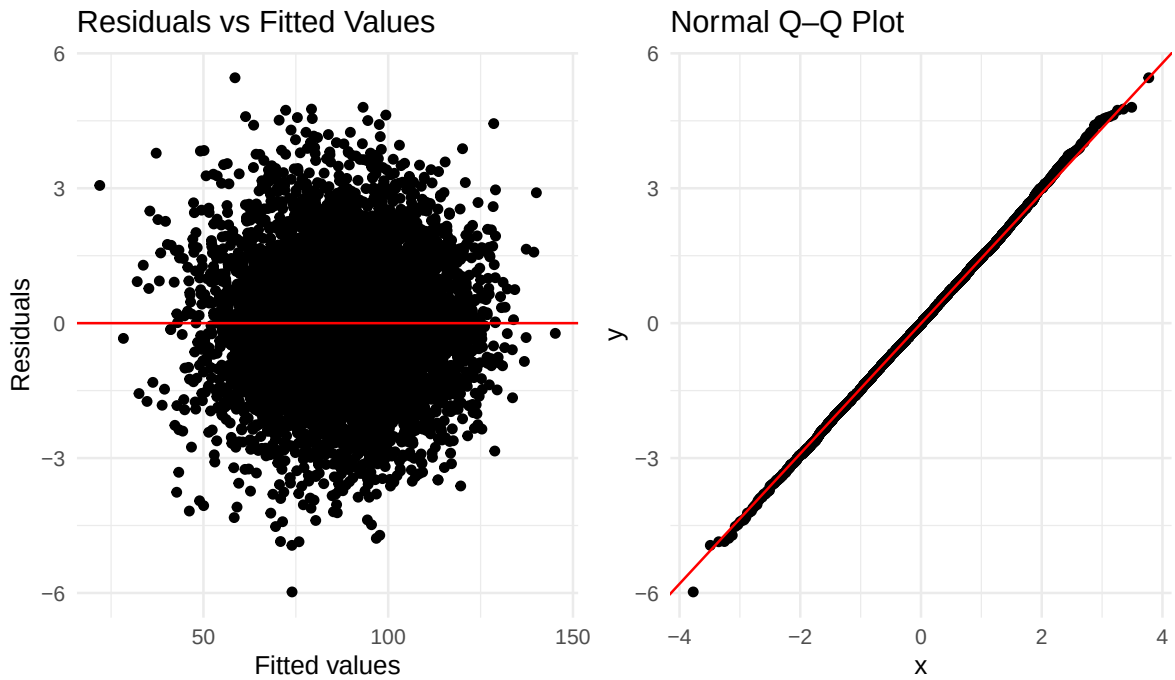

Figure 8: Residual Diagnostics: Residuals vs Fitted (left) and Normal Q–Q Plot (right).

Residuals vs. Fitted Values Plot Figure @ref(fig:q5_residual_diagnostics) (left) assesses whether residuals are randomly distributed around zero, which reflects the assumption of constant variability ("homoscedasticity"). Residuals are spread evenly around zero with no funnel shape or curvature. This indicates variance is stable across time, the model does not systematically under- or over-predict symptoms and the linearity assumption is appropriate. The model fits the data well and the variability is appropriately captured.

Distribution of Patient-Specific Intercepts and Slopes To evaluate whether the random-effects model appropriately captured differences between patients, we examined the distribution of the estimated subject-specific intercepts (patients' baseline BPRS

scores) and subject-specific slopes (their rates of symptom change over time) in Figure @ref(fig:q5_random_effects_diagnostics). The random intercepts (left Q–Q plot) follow the theoretical normal line very closely, with only minor deviations at the extreme tails. The random slopes (middle Q–Q plot) also align closely with the diagonal reference line, supporting approximate normality. The scatterplot of random intercepts versus random slopes (right) shows no strong linear relationship, implying that starting severity does not systematically predict a patient's rate of change.

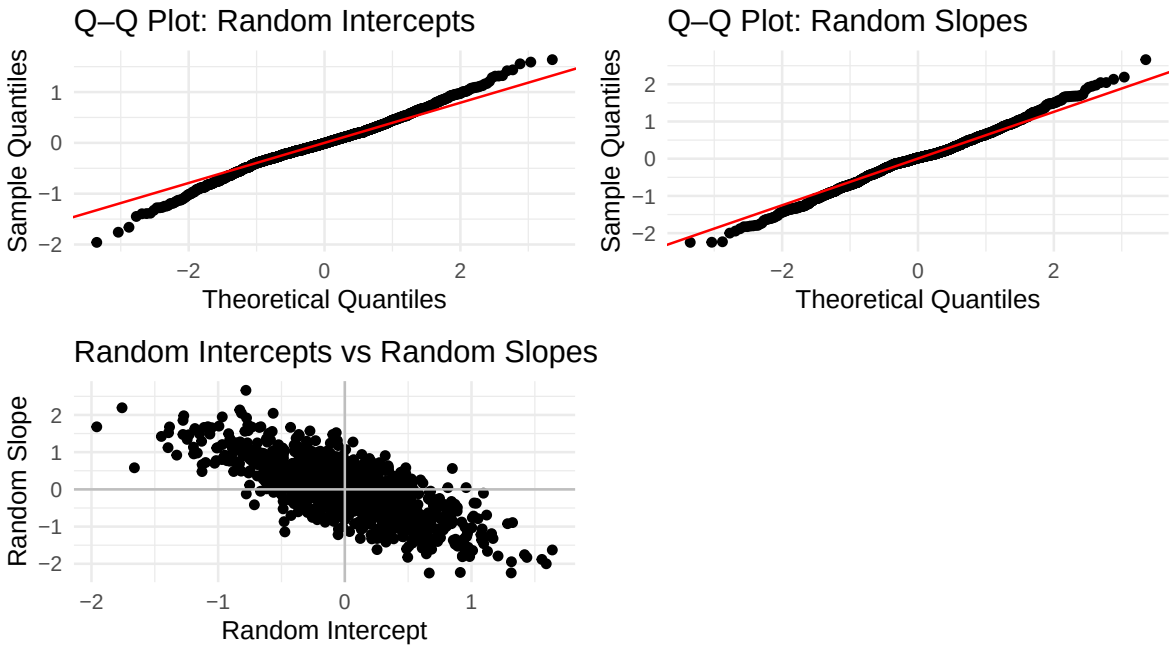

Figure 9: Diagnostics of Random Effects:
Q–Q plots for random intercepts (left) and random slopes (middle), and their joint distribution (right).

Comparison With the Two-Stage Analysis

To better understand how each patient's symptoms (BPRS scores) changed over time, we estimated individual starting points (intercepts) and rates of change (slopes) using two different approaches. First, in Question 4, we applied a two-stage method, fitting a separate linear regression for each patient to obtain individual intercepts and slopes. Second, we used a random-effects mixed model, which estimates all patient-specific intercepts and slopes simultaneously. We then compared the two methods to see whether the advanced mixed model provides meaningful improvement.

To compare the two-stage approach with the mixed-effects model, we examined patient-specific intercepts and slopes estimated by each method. The intercepts showed very strong agreement

between the two approaches (Figure @ref(fig:q5_two_stage_vs_mixed), left), with a correlation of r = 0.94 and points closely following the 45-degree line. In contrast, the agreement for slopes was moderate (Figure @ref(fig:q5_two_stage_vs_mixed), right; r = 0.77), and the two-stage estimates displayed greater variability and several extreme values. This pattern reflects the inherent instability of estimating individual change rates from limited per-patient data. The mixed-effects model, by using all available data simultaneously, shrinks extreme slopes toward the population mean and produces more stable and reliable estimates.

Overall, the two methods give broadly similar patient-specific results, but the mixed model provides a clearer and more stable representation of individual symptom trajectories, particularly for the rate of change over time

Subject-specific intercepts and slopes from mixed model vs two-stage estimates

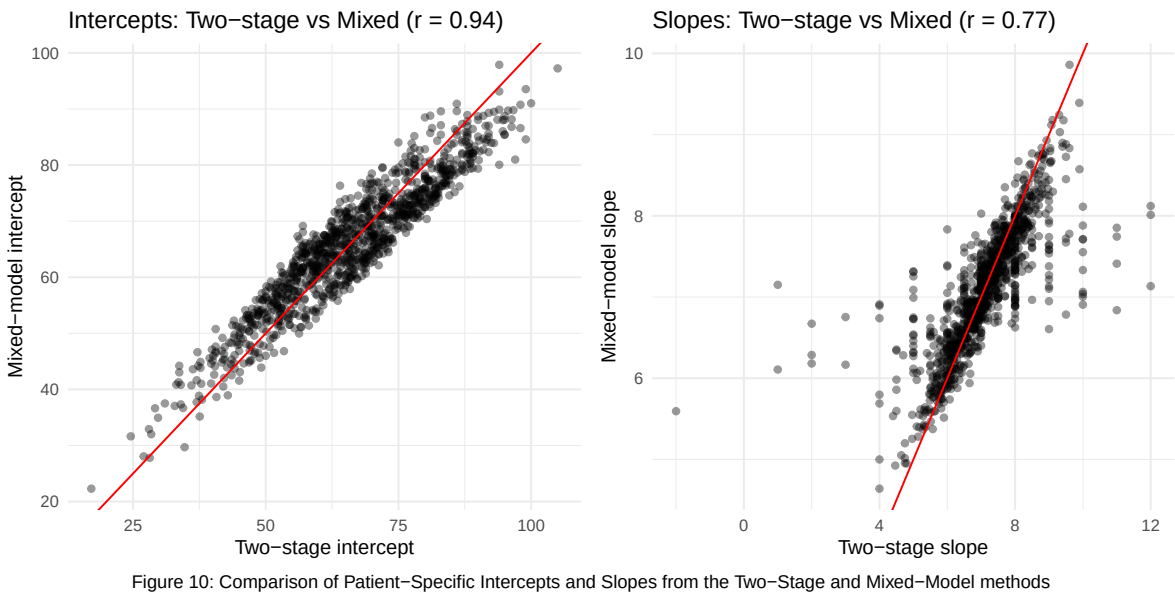

Figure 8: Patient-specific intercepts (left) and slopes (right): two-stage vs mixed-model estimates.

Why are the mixed-model estimates still preferred?

Compared to the two-stage approach, mixed-effects models are preferred because they correctly account for the correlation among repeated measurements within each patient, whereas the two-stage method treats each patient's data independently. Mixed models also allow a flexible covariance structure for intercepts and slopes, leading to more accurate uncertainty estimates. In contrast, the two-stage approach produces unstable slopes when patients have few

observations and cannot model within-patient correlation. Overall, the mixed model provides more reliable and statistically valid patient-specific estimates.

Results

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: fm_adj
   Data: alzheimer_long

REML criterion at convergence: 27121.3

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.6390 -0.5982 -0.0086  0.5904  3.3222

Random effects:
 Groups   Name        Variance Std.Dev. Corr
 patid    (Intercept) 0.8067   0.8982
          time        0.6863   0.8284   -0.61
 trial    (Intercept) 8.0037   2.8291
 Residual             2.6980   1.6426
Number of obs: 6220, groups:  patid, 1253; trial, 25

Fixed effects:
                  Estimate Std. Error         df t value Pr(>|t|)
(Intercept)     -6.029e+01  4.169e+00  9.534e+02 -14.461  < 2e-16 ***
time             7.220e+00  4.026e-02  9.957e+02 179.318  < 2e-16 ***
sexFemale       -1.609e-01  8.760e-02  1.169e+03  -1.837   0.0665 .
age              1.697e+00  4.006e-02  1.044e+03  42.364  < 2e-16 ***
bmi              1.784e-01  2.729e-02  1.204e+03   6.537 9.26e-11 ***
jobPaid Job     -5.136e+00  8.042e-01  1.059e+03  -6.386 2.54e-10 ***
adl              6.535e-02  1.178e-01  1.054e+03   0.555   0.5791
wzcNursing Home  1.915e+00  9.790e-02  1.258e+03  19.565  < 2e-16 ***
cdrsb0           2.002e-03  7.603e-03  1.083e+03   0.263   0.7924
time:cdrsb0     -2.178e-02  4.149e-03  9.707e+02  -5.248 1.88e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Correlation of Fixed Effects:
            (Intr) time   sexFml age    bmi    jbPdJb adl    wzcNrH cdrsb0
time        -0.009
sexFemale    0.026 -0.006
age         -0.981 -0.003 -0.054
bmi         -0.773 -0.005 -0.002  0.693
jobPaid Job  0.955 -0.001  0.025 -0.960 -0.697
adl         -0.975  0.002 -0.001  0.980  0.708 -0.983
wzcNursngHm  0.052 -0.016  0.020 -0.086 -0.003  0.037 -0.009
cdrsb0      -0.017  0.444  0.044  0.006 -0.014 -0.013  0.011  0.016
time:cdrsb0  0.006 -0.677  0.005  0.002  0.003  0.000  0.001  0.000 -0.651
```

Table 10: Fixed effects estimates from mixed model

effect	term	estimate	std.error	statistic	df	p.value
fixed	(Intercept)	-60.293	4.169	-14.461	953.390	0.000
fixed	time	7.220	0.040	179.318	995.729	0.000
fixed	sexFemale	-0.161	0.088	-1.837	1169.059	0.067
fixed	age	1.697	0.040	42.364	1043.954	0.000
fixed	bmi	0.178	0.027	6.537	1204.048	0.000
fixed	jobPaid Job	-5.136	0.804	-6.386	1058.656	0.000
fixed	adl	0.065	0.118	0.555	1053.977	0.579
fixed	wzcNursing Home	1.915	0.098	19.565	1258.449	0.000
fixed	cdrsb0	0.002	0.008	0.263	1082.728	0.792
fixed	time:cdrsb0	-0.022	0.004	-5.248	970.693	0.000

Table 11: Random-effects variance–covariance components

Group	Term1	Term2	Value	SD_or_Corr
patid	(Intercept)	NA	0.807	0.898
patid	time	NA	0.686	0.828
patid	(Intercept)	time	-0.455	-0.611
trial	(Intercept)	NA	8.004	2.829
Residual	NA	NA	2.698	1.643

The linear mixed-effects model was fitted to 6,220 observations from 1,253 patients across

25 trials. The REML criterion at convergence was 27,121.3, and scaled residuals ranged from −3.64 to 3.32.The random-effects structure included patient-level random intercepts and random slopes for time. These patient-specific intercepts and slopes were correlated. The model also included trial-level random intercepts to account for between-trial variation. The full set of estimated variances, standard deviations, and correlations for all random-effect terms is presented in Table 5.

Several fixed effects were statistically significant, including the intercept, time, age, BMI, job status, WZC category, and the time-by-cdrsb0 interaction, all with ($p<0.001$). The effects of sex, ADL, and the main effect of cdrsb0 were not statistically significant. Full estimates, standard errors, and significance levels for all fixed effects are presented in Table 4.

Discussion

Using a mixed-effects model, we examined how psychiatric symptoms change over time in patients with Alzheimer's disease. This modelling approach accounts for individual differences in both baseline symptom severity and rates of progression.

The analysis showed that psychiatric symptoms worsen steadily over time, with patients on average increasing by about 7 BPRS points per year. However, not all patients start at the same level or decline at the same speed. Our model captured this variation and revealed important patient characteristics that help explain these differences.

Patients who were older, had higher BMI, poorer daily functioning (ADL), lived in a nursing home, or did not have a paid job tended to have more severe psychiatric symptoms at baseline. In addition, patients with more impaired cognition at diagnosis (higher CDR-SB) showed faster worsening of psychiatric symptoms over time.

Overall, this approach provides a detailed picture of how psychiatric symptoms evolve in Alzheimer's disease. It shows which baseline factors are linked with faster decline, and it gives individualized estimates of symptom progression. These findings can help to better understand patient variability, identify patients who may be at higher risk for rapid deterioration, and guide monitoring and care strategies over the course of the disease.

Reflection on parameterization

The chosen parameterization had two main components:

Mean structure (fixed effects): We specified a linear effect of time, adjusted for sex, age, BMI, job status, ADL, nursing home residence (WZC), baseline CDR-SB, amyloid PET, and the interaction between time and CDRSB. This reflects a clinical hypothesis that:

1) symptom severity depends on demographic, functional, and disease severity markers at baseline, and

2) cognitive status modifies the trajectory of psychiatric symptoms (time $\times$ CDRSB0).

Although the assumption of linear change simplifies interpretation, non-linear patterns could be explored in future work if justified..

For the covariance structure, we selected patient-level random intercepts and random slopes for time with an unstructured covariance matrix. This parameterization captures individual differences in baseline severity and rates of change while allowing them to be correlated. Trial-level random intercepts accounted for between-trial variation. Overall, this structure provides a good balance between flexibility and interpretability. Alternative parameterizations, such as non-linear time or more complex random effects, could be considered, but the chosen model is appropriate and coherent for the current analysis.

Recommendations for future similar experiments

From a modeling and design perspective, these are recommendations that emerged:

1) systematic recording of dropout reasons: Since dropout can bias longitudinal analyses, future studies should routinely document why patients leave the study (death, institutionalization, refusal, health decline). This would allow sensitivity analyses under different missing-data assumptions (MAR vs MNAR).

2) Richer covariate information on treatment and care: Including medication use (e.g., antipsychotics, antidepressants), psychosocial interventions, caregiver burden, and comorbidities might help explain additional between-patient variation and could identify modifiable risk factors for rapid deterioration.

3) Parallel assessment of multiple outcome domains Since cognition, function, and psychiatric symptoms are closely related in Alzheimer's disease, future experiments could be designed to jointly model these domains rather than analyzing each in isolation. This would provide a more integrated picture of disease progression and may better inform clinical decision-making.